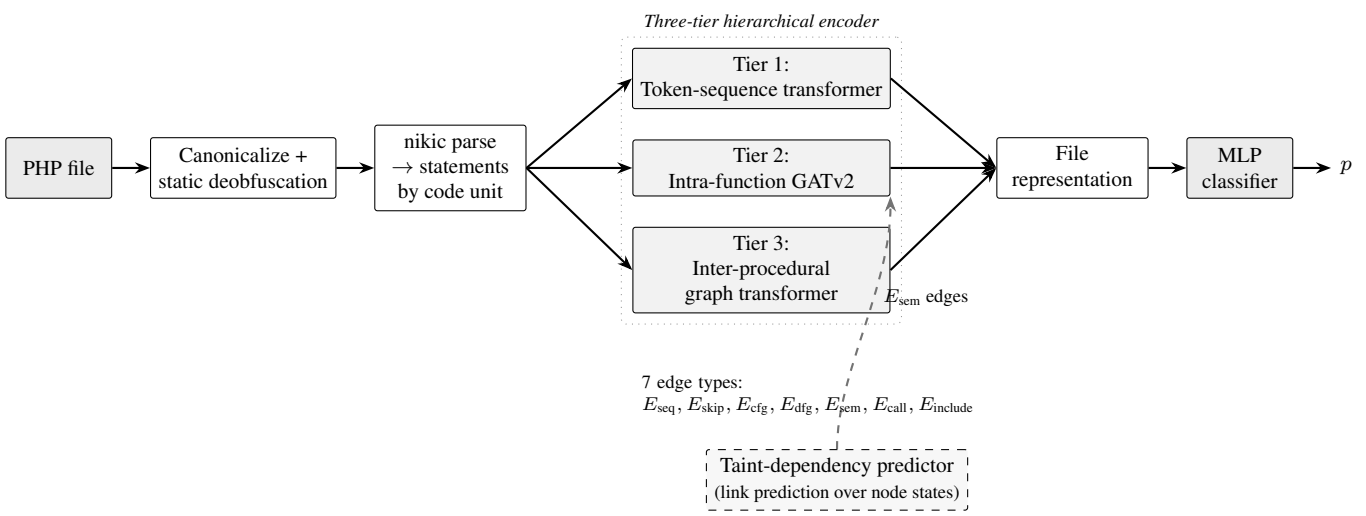
Three-tier hierarchical encoder
PHP file
Canonicalize + static deobfuscation
nikic parse → statements by code unit
Tier 1: Token-sequence transformer
Tier 2: Intra-function GATv2
Tier 3: Inter-procedural graph transformer
File representation
MLP classifier
p
E_{sem} edges
7 edge types:
E_{seq}, E_{skip}, E_{cfg}, E_{dfg}, E_{sem}, E_{call}, E_{include}
Taint-dependency predictor
(link prediction over node states)